

THE LYMAN BANDS OF MOLECULAR HYDROGEN¹

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ABSTRACT

The Lyman bands of H_2 have been investigated under high resolution with a view to improving the rotational and vibrational constants of H_2 in its ground state. Precise B_v and ΔG values have been obtained for all vibrational levels of the ground state. One or two of the highest rotational levels of the last vibrational level ($v = 14$) lie above the dissociation limit. Both the B_v'' and $\Delta G''$ curves have a point of inflection at about $v'' = 3$. This makes it difficult to represent the whole course of each of these curves by a single formula and therefore makes the resulting equilibrium constants somewhat uncertain. This uncertainty is not very great for the rotational constants for which we find

$$B_e = 60.864 \quad \alpha_e = 3.0764 \text{ cm}^{-1}$$

but is considerable for the vibrational constants ω_e and $\omega_e x_e$ for which three-, four-, five-, and six-term formulae give results diverging by $\pm 1 \text{ cm}^{-1}$. The rotational and vibrational constants for the upper state ($B^1\Sigma_u^+$) of the Lyman bands are also determined. An appreciable correction to the position of the upper state is found.

1. INTRODUCTION

The most precise values at present available for the rotational and vibrational constants of the hydrogen molecule in its ground electronic state are those derived from the quadrupole spectrum in the photographic infrared (Herzberg 1950a) and from Raman data (Stoicheff 1957). However, these constants represent precisely only the lowest vibrational and rotational levels, up to $v = 3$ and $J = 5$. For the higher vibrational and rotational levels one is dependent on the emission spectrum of H_2 in the vacuum ultraviolet. The frequency dispersion in the vacuum ultraviolet is low, and therefore, in the early work of Hyman (1930) and Jeppesen (1933), vibrational and rotational differences were obtained only with rather modest wave number accuracy. As a result there were differences of up to 4 cm^{-1} between the predicted and observed wave numbers of the quadrupole infrared bands.

Recent improvements in vacuum ultraviolet spectroscopy have made it possible to measure lines in this region with much greater accuracy. Because of the importance of the H_2 molecule in general and of a precise knowledge of all its energy levels in particular, it was considered to be of interest to remeasure a sufficient part of the far ultraviolet spectrum in order to obtain the constants of all vibrational levels of H_2 in the ground state. The behavior of the levels near the dissociation limit promised to be of particular interest (compare the early work of Beutler 1934).

There are two well-known emission band systems of H_2 in the vacuum ultraviolet; the Lyman bands ($B-X$) representing a $^1\Sigma_u^+ \rightarrow ^1\Sigma_g^+$ transition and the Werner bands ($C-X$) representing a $^1\Pi_u \rightarrow ^1\Sigma_g^+$ transition. Since the former

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PLATE I

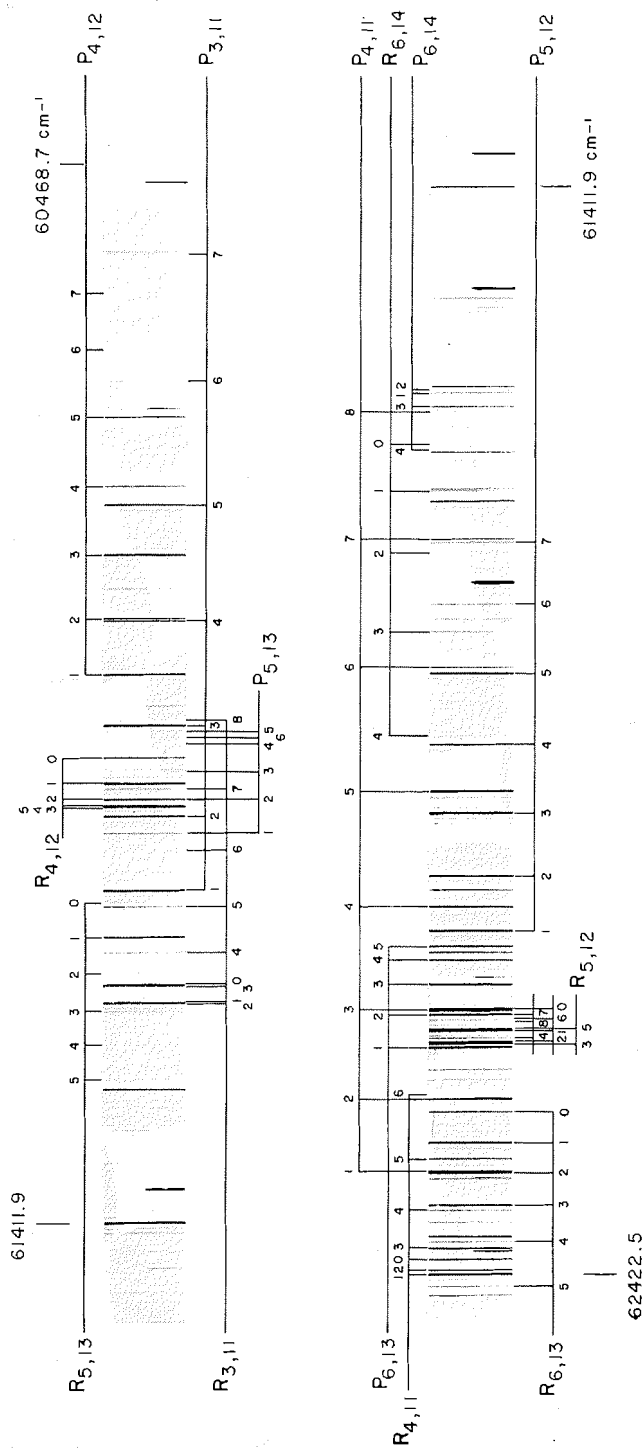


FIG. 1. Section of the spectrum of H₂ extending from 1655 to 1600 Å. Some of the Lyman bands are indicated.

lie at longer wavelengths and therefore allow one to obtain a higher wave number accuracy they were chosen for the present work. Moreover, the higher vibrational levels of the ground state are more favored in this transition than in the Werner bands on account of the Franck-Condon principle.

2. EXPERIMENTAL

The Lyman bands of H_2 can be produced in any discharge through hydrogen. In the present work most spectra were obtained by means of an electrodeless discharge tube operated by a high-frequency oscillator (a Raytheon microtherm unit). In order to bring out lines of high rotational quantum number a few spectra were taken with a high-current tube with electrodes of conventional type. The electrodeless tube was also used in some experiments with liquid nitrogen cooling in order to reduce the Doppler width of the lines and to simplify the spectrum by the exclusion of lines with high J .

The spectra were taken in the fourth and fifth orders of our 3-meter vacuum spectrograph, that is with a dispersion of 0.66 or 0.5 Å/mm. The orders were separated by means of a foregrating as described by Douglas and Herzberg (1957). Standards were supplied by a Fe hollow cathode discharge tube. The light from this source was sent through the H_2 discharge tube onto the slit of the spectrograph. The Fe lines were photographed in second order. No attempt to eliminate the error introduced by the method of overlapping orders was made, that is the absolute accuracy of the wavelengths may only be ± 0.01 Å corresponding to ± 0.5 cm⁻¹ but the relative accuracy (and therefore the accuracy of rotational and vibrational differences) was ± 0.05 cm⁻¹ for unblended lines. Lines that were more than 0.5 cm⁻¹ apart were resolved, that is the attained resolving power was well above 100,000. The measurements were made with a photoelectric comparator built in this laboratory by Dr. B. P. Stoicheff according to the design of Tomkins and Fred (1951).

In Fig. 1 a section of the long wavelength end of the spectrum is reproduced showing some of the bands with the highest v'' values.

3. ANALYSIS OF THE SPECTRUM

The Lyman bands represent a typical many-line spectrum (see Fig. 1); individual bands or branches are not at all obvious. However, with the bands analyzed by Hyman and Jeppesen, it was relatively easy to proceed and find other bands. Some difficulty was encountered in assigning the lines of the bands of highest v'' since, because of the high D'' values, the form of the branches is rather unusual and since in addition the branches are very short. However, the use of combination relations has in each case made a reliable analysis possible.

In Table I are listed the wave numbers of the lines of all the bands that have been analyzed. Lines used more than once are indicated by asterisks. There are a considerable number of lines which remain unidentified. These are listed in Table II for the region 1685–1380 Å. It is very difficult to see what their identification may be unless they are "extra lines" of some of the weaker bands with perturbations.

TABLE I
Vacuum wave numbers of the lines of the Lyman bands of H₂

0-3 band $\nu_0 = 78420.9_s \text{ cm}^{-1}$		0-4 band $\nu_0 = 74952.9_s \text{ cm}^{-1}$		0-5 band $\nu_0 = 71711.4_2 \text{ cm}^{-1}$		0-6 band $\nu_0 = 68697.6_2 \text{ cm}^{-1}$		0-7 band $\nu_0 = 65915.5_1 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		78459.84		74992.03		71750.37		68736.57		65953.59*
1	78319.93	436.16	74857.76	973.88	71621.79	737.96*	68613.71	729.85	65837.40	953.59*
2	157.67	349.94	706.60	898.97	481.84	674.29	485.32	677.80	720.83	913.13
3	77935.62	202.32	501.38	768.14	293.77	560.40	314.14	580.90*	567.13	833.90
4	655.82	77994.53	243.96	582.66	058.70	397.40	101.51*	440.50	378.21	716.92
5	320.81	728.34	73936.79	344.51	70779.47	187.06	67850.78	258.44	156.27	563.88
6			582.91	055.89	459.80	70931.58	563.71*	036.99*	64903.86	
7			185.21	73719.74	099.09	633.66	243.13	67777.74	623.79	
8			72747.19	336.30	69704.27	296.16†				
9			272.22	72916.92	271.89†	69922.21				
10			71764.28	454.32						
11			225.96							

0-8 band $\nu_0 = 63372.3_7 \text{ cm}^{-1}$		1-3 band $\nu_0 = 79739.4_0 \text{ cm}^{-1}$		1-4 band $\nu_0 = 76271.3_9 \text{ cm}^{-1}$		1-5 band $\nu_0 = 73029.8_3 \text{ cm}^{-1}$		1-6 band $\nu_0 = 70016.1_0 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0				79776.47		76308.15		73066.80		70052.99
1	63300.32*	63416.49	79638.50	749.13	76175.98	286.36	72940.14	050.49	69932.01	042.48
2	195.99	388.63	474.13	657.09	022.92	205.68	798.37		801.77	69984.63
3	060.76	327.55	248.19	502.17†	75813.85	067.78	606.27	72860.21	626.72	880.53
4	62896.43	235.35*	78962.84	285.79	550.97*	75873.52	365.68		408.76	731.16
5	704.70	112.42	620.79		236.56	625.36	079.10		150.47	539.65
6			227.84						68854.47	306.48
7					74466.05	74976.15	71380.09		524.11	035.63

1-7 band $\nu_0 = 67233.9_2 \text{ cm}^{-1}$		1-8 band $\nu_0 = 64690.7_8 \text{ cm}^{-1}$		1-9 band $\nu_0 = 62397.8_2 \text{ cm}^{-1}$		1-10 band $\nu_0 = 60371.5_6 \text{ cm}^{-1}$		2-3 band $\nu_0 = 81020.8_3 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		67270.80		64727.89		62434.14		60408.40		
1	67155.85	266.16	64618.70	729.02	62332.28	442.67	60312.92	423.30	80919.91	81025.36
2	037.36*	220.13	512.46	695.33	239.04	422.54*	233.84	416.57		80928.63
3	66879.80	133.60	373.30	627.16	119.34	373.22	135.27	389.14	524.79	
4	685.20	007.89	203.20	525.85	61975.07	298.41*	019.41	342.11		
5	455.94	66844.81	004.28	393.25	808.77	197.94	59888.56	277.45		
6	194.78	646.66	63779.54	231.52	622.98		745.54	197.57		
7	65904.64*	416.10	531.82	043.40	420.89	61930.33	593.42	105.10		
8	588.72	156.21	266.71†	63831.89†			435.51			
9	250.44	65870.02								
10	64890.37									

2-4 band $\nu_0 = 77552.8_2 \text{ cm}^{-1}$		2-5 band $\nu_0 = 74311.2_3 \text{ cm}^{-1}$		2-6 band $\nu_0 = 71297.5_3 \text{ cm}^{-1}$		2-7 band $\nu_0 = 68515.3_3 \text{ cm}^{-1}$		2-8 band $\nu_0 = 65972.2_1 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		77588.08		74346.49		71332.85				66007.47
1	77457.48	563.14*	74221.53	327.12	71213.56	319.06	68437.24*	68542.70	65900.21	005.74
2	302.70	477.65	078.10*	253.02	081.62	256.49	317.02	491.10*	792.24	65967.17
3	090.45	333.02*	73882.73	125.56	70903.32	146.17	156.38		649.85	892.71
4	76822.57	131.76	637.34	73946.19	680.63	70989.51	67956.99		474.93	783.86
5	502.03	76874.63	344.59	717.16	416.14	788.76	721.58		269.92	642.58
6	131.92	565.26	007.56	440.99	112.61	546.20	452.85		037.68	471.11
7	75715.46		72629.38	120.51	69773.44	268.10			64781.24*	272.37
8	256.63								504.08	

*Blended line.
†Doubtful.

TABLE I (Continued)
 Vacuum wave numbers of the lines of the Lyman bands of H₂

2-9 band $\nu_0 = 63679.2_6 \text{ cm}^{-1}$			2-10 band $\nu_0 = 61652.9_9 \text{ cm}^{-1}$		2-11 band $\nu_0 = 59916.3_1 \text{ cm}^{-1}$		3-3 band $\nu_0 = 82267.6_0 \text{ cm}^{-1}$		3-4 band $\nu_0 = 78799.5_9 \text{ cm}^{-1}$	
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		63714.45		61688.28*		59951.42				78833.48
1	63613.78	719.20	61594.39	699.82	59865.51	970.98		82267.90		805.41
2	518.82	693.66	513.62	688.28*	800.39	975.28			78548.18	715.86
3	395.94	638.74	411.92	654.66	722.29	965.15	81767.27		332.90	566.00*
4	247.00*	555.90	291.23	600.08	633.41	942.13			061.06*	
5	074.50	447.10	154.20*	526.82	536.04	908.54			77734.66	092.80
6	62881.22	314.92	003.63	437.12					357.78	77773.77
7	670.13	161.32	60842.79	334.15					76933.83	406.50
8	445.08		675.61						465.72	
9									75958.74	76527.53

3-5 band $\nu_0 = 75558.0_5 \text{ cm}^{-1}$			3-6 band $\nu_0 = 72544.3_0 \text{ cm}^{-1}$		3-7 band $\nu_0 = 69762.1_2 \text{ cm}^{-1}$		3-8 band $\nu_0 = 67218.9_3 \text{ cm}^{-1}$		3-9 band $\nu_0 = 64926.0_2 \text{ cm}^{-1}$	
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		75591.80		72578.05		69795.95		67253.20		64959.81
1	75468.26	569.40	72460.04	561.40	69683.97	785.14	67146.88	248.08	64860.47	961.59
2	323.45	491.12	326.93	494.58	562.35	730.10	037.36*	205.10	764.08	932.02
3	125.17	358.19	145.71	378.75	398.74	631.86	66892.20	125.29	638.30	871.35
4	74875.53	172.13	71918.73		195.10	491.87	713.08	009.50	485.16	781.24*
5	577.23	74935.06	648.41*		68954.18	312.22	502.46	66860.43	307.08	665.09
6	233.59	650.26	338.49		678.28*	095.71	263.90		106.99	521.97*
7	73847.25	320.77	70991.45		371.92	68844.78	65994.66	466.83	63888.28	357.10
8					036.99*				654.33	
9					67679.26*					

3-10 band $\nu_0 = 62899.7_6 \text{ cm}^{-1}$			3-11 band $\nu_0 = 61163.1_0 \text{ cm}^{-1}$		3-12 band $\nu_0 = 59748.1_2 \text{ cm}^{-1}$		4-3 band $\nu_0 = 83480.5_8 \text{ cm}^{-1}$		4-4 band $\nu_0 = 80012.5_7 \text{ cm}^{-1}$	
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		62933.49		61197.18		59781.65				80045.25
1	62841.18	942.35	61112.21*	213.41*	59706.64	807.34	83379.66	83476.86	79917.37	014.55
2	758.88	926.59*	045.70	213.41*				210.94	759.88	79921.12
3	654.25	887.35	60964.73	197.70			82976.29		542.05	
4	529.42	826.05	871.46	168.04					266.09	
5	386.83	744.82	768.68	126.71					78934.86	279.57
6	229.45	645.91	659.62	076.57						
7	060.96	533.92	548.21	021.17	59395.70				120.22	78576.07
8	61884.73	411.49	438.78	60959.67						
9	705.63	281.07	336.50							
10			247.30							
11			179.44							

4-5 band $\nu_0 = 76771.0_1 \text{ cm}^{-1}$			4-6 band $\nu_0 = 73757.2_3 \text{ cm}^{-1}$		4-7 band $\nu_0 = 70975.1_0 \text{ cm}^{-1}$		4-8 band $\nu_0 = 68431.9_6 \text{ cm}^{-1}$		4-9 band $\nu_0 = 66139.0_6 \text{ cm}^{-1}$	
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		76803.19		73789.64		71007.65		68464.40		66170.87*
1	76681.11	778.28	73673.16	770.36	70897.06	70994.22	68359.94	457.12	66073.55	170.87*
2	534.70		538.46	699.66	773.97	935.30	240.26	410.50	65975.81*	137.05
3	334.15		354.68	578.70	607.85	831.94	101.51*	325.46	847.37*	071.51
4	080.61		123.83	409.18	400.39	685.94	67918.29	203.68	690.29	65975.81*
5	75777.17		72848.81	193.34	154.27	498.94	702.71	047.32	507.14	851.81
6	427.21		532.36	72931.78*	69872.46		457.42	67856.51	300.85	702.38
7	034.22		178.21				185.98		074.88	530.06*
8			71789.89†							

*Blended line.

†Doubtful.

TABLE I (Continued)
Vacuum wave numbers of the lines of the Lyman bands of H₂

4-10 band $\nu_0 = 64112.7_4 \text{ cm}^{-1}$			4-11 band $\nu_0 = 62376.0_8 \text{ cm}^{-1}$		4-12 band $\nu_0 = 60961.1_0 \text{ cm}^{-1}$		4-13 band $\nu_0 = 59911.9_2 \text{ cm}^{-1}$		5-3 band $\nu_0 = 84660.6_0 \text{ cm}^{-1}$	
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		64145.09		62408.42		60993.38		59944.07		84670.99
1	64054.17	151.38	62325.33	422.54*	60919.21	61016.44	59880.73	977.98	84559.75	653.23
2	63970.54	131.82	257.41	418.62	869.32	030.59*	852.37	60013.80		
3	863.34	087.52	173.79	397.92	812.92	037.02*	828.97	053.14	152.70	
4	734.57	019.98	076.09	362.10	752.48	037.02*	813.24	098.64		
5	586.88*	63931.54	61968.80	313.50	691.13	036.08	809.42	154.15		
6	423.41		853.58	252.26	632.79		823.11	224.73		
7			734.86		582.38		862.13	317.92		
8			617.42		540.24		935.72			

5-4 band $\nu_0 = 81192.5_9 \text{ cm}^{-1}$			5-5 band $\nu_0 = 77951.0_3 \text{ cm}^{-1}$		5-6 band $\nu_0 = 74937.3_0 \text{ cm}^{-1}$		5-7 band $\nu_0 = 72155.1_2 \text{ cm}^{-1}$		5-8 band $\nu_0 = 69611.9_8 \text{ cm}^{-1}$	
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0				77982.35		74968.39		72186.59*		69643.23
1	81097.32	81190.90	77861.32	954.87	74853.21	946.78	72077.00	170.56	69539.93	633.51*
2	80916.90		713.92	870.15	717.30	872.46	71952.81	108.03	427.99	583.20
3	718.38		510.46	726.27*	530.99	746.85	784.12	71999.93	277.61	493.42
4			254.00	528.28	296.54	571.53	573.02		091.03	365.95
5			76945.25	277.39		016.69	348.82	322.36	68870.71*	202.79
6			589.33		73694.40	081.80	034.97		619.65	006.09*
7			189.52	76630.97		73773.24	70714.27		341.54	68781.18
8			75749.36†						039.55	529.30

5-9 band $\nu_0 = 67319.0_2 \text{ cm}^{-1}$			5-10 band $\nu_0 = 65292.7_6 \text{ cm}^{-1}$		5-11 band $\nu_0 = 63556.1_0 \text{ cm}^{-1}$		5-12 band $\nu_0 = 62141.1_2 \text{ cm}^{-1}$		5-13 band $\nu_0 = 61091.9_2 \text{ cm}^{-1}$	
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		67350.12		65323.84		63586.88*		62172.33		61122.77
1	67253.20*	346.92	65234.12	327.62	63505.27	598.81	62099.21	192.77	61060.84	154.20*
2			149.28	304.50	435.88*	591.59	048.11	203.32	030.59	186.46
3	023.72		039.65	255.39	350.13	565.91	61989.24	205.05	005.29	221.16
4			64907.34	182.15	249.41	524.14	925.15	200.09		260.94
5	66675.14		754.84	087.05	136.84	468.90	859.06	191.50	60977.49	
6	462.93		585.60	64972.95	015.93		794.98	182.48	982.68	
7	230.45		403.09	842.90	62890.55		738.10	178.75		
8			209.52		765.40		693.55	183.83		
9							669.63			

5-14 band $\nu_0 = 60469.9_8 \text{ cm}^{-1}$			6-5 band $\nu_0 = 79098.6_1 \text{ cm}^{-1}$		6-6 band $\nu_0 = 76084.8_8 \text{ cm}^{-1}$		6-7 band $\nu_0 = 73302.7_0 \text{ cm}^{-1}$		6-8 band $\nu_0 = 70759.5_0 \text{ cm}^{-1}$	
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		60501.01		79128.74			73332.70		70790.97	
1	60452.52	546.10	79008.94	098.94		76090.53	73224.40	314.48	70687.30	777.94
2	451.32	606.30	78860.32		75863.87		099.16	248.56		723.84
3	468.70	684.49	654.62	78862.49	675.49	75882.87	72928.23	136.00	421.91	629.65
4	511.40	786.31		394.08			713.74	72978.54		
5			081.56	401.88			458.51	779.40		
6			77720.17				165.64			
7			314.05				71838.59	262.59		

*Blended line.

†Doubtful.

TABLE I (Continued)
Vacuum wave numbers of the lines of the Lyman bands of H₂

6-9 band $\nu_0 = 68466.6_0 \text{ cm}^{-1}$		6-10 band $\nu_0 = 66440.3_4 \text{ cm}^{-1}$		6-11 band $\nu_0 = 64703.6_8 \text{ cm}^{-1}$		6-12 band $\nu_0 = 63288.7_0 \text{ cm}^{-1}$		6-13 band $\nu_0 = 62239.5_2 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		68496.63				64733.51		63318.53		62269.34
1	68401.00*	491.10*	66381.52	66471.78	64652.80	742.85	63247.00*	336.80	62208.32	298.41*
2	300.93	450.40		445.04	582.50	732.00	194.42	343.94	177.69	327.12
3	167.78	375.75*			494.17	702.04	133.30	341.23	149.44	357.31
4	003.68	268.71			389.99	654.94	065.83	330.77	126.68	391.78
5	67811.37	131.75		211.43	272.98	593.35	62995.38	315.60	113.66*	433.03
6	593.94				146.52	521.97*	926.59*	300.32*	113.66*	
7	354.73	67777.74			014.86	438.81	862.35	286.89		
8	098.01						810.88			
9							778.93			

6-14 band $\nu_0 = 61617.5_6 \text{ cm}^{-1}$		7-5 band $\nu_0 = 80213.9_7 \text{ cm}^{-1}$		7-6 band $\nu_0 = 77200.2_4 \text{ cm}^{-1}$		7-7 band $\nu_0 = 74418.0_6 \text{ cm}^{-1}$		7-8 band $\nu_0 = 71874.9_2 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		61647.08		80243.15		77229.00		74446.92		71903.82
1	61600.08*	690.06	80124.38	211.00	77116.12	202.71	74339.87	426.52	71802.90	889.53*
2	597.46	747.14	79974.71	118.49	76977.79	121.67	74339.87	357.25	688.63	832.55
3	612.74	820.58	766.66	79967.05	786.93*	76987.25	040.01	240.38	533.71	733.79
4	651.98	916.98		502.17*	545.94	801.15	73822.81*	078.10*	340.41	
5			78986.04		257.31	565.94	562.68*	73871.87	111.32	
6					75924.49		264.79	625.73*	70849.67	
7					550.97*		72931.78*	342.62	559.06	

7-9 band $\nu_0 = 69581.9_6 \text{ cm}^{-1}$		7-10 band $\nu_0 = 67555.7_0 \text{ cm}^{-1}$		7-11 band $\nu_0 = 65819.0_4 \text{ cm}^{-1}$		7-12 band $\nu_0 = 64404.0_6 \text{ cm}^{-1}$		7-13 band $\nu_0 = 63354.8_8 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		69610.88				65847.37*		64432.76		63383.62
1	69516.45	603.12	67497.09	67583.86	65768.21	854.82	64362.17	448.74	63323.80	410.40
2	415.19	559.05	409.97	553.86	696.65	840.58	308.65	452.60	291.98	435.88
3	279.75	480.09	295.73	496.33*	606.13	806.44	245.31	445.61	261.40	461.71
4	112.38	367.99		* 412.13		754.19	174.56	430.10	235.35*	490.86
5	68915.81	224.92	66995.50	304.65		686.54	099.76	408.88	218.25	527.21
6			815.55						212.94	576.18
7	448.08		620.51	031.67					230.22	

7-14 band $\nu_0 = 62732.9_2 \text{ cm}^{-1}$		8-6 band $\nu_0 = 78284.2_6 \text{ cm}^{-1}$		8-7 band $\nu_0 = 75502.0_8 \text{ cm}^{-1}$		8-8 band $\nu_0 = 72958.9_4 \text{ cm}^{-1}$		8-10 band $\nu_0 = 68639.7_2 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		62761.61		78312.24		75529.89		72986.90		68667.61
1	62715.60	802.21	78200.19*	284.08	75423.78	507.66	72886.86	970.83	68580.90*	665.07
2	711.61	855.68	061.06*	200.19*	296.58	435.83	771.70	911.22	492.92	632.49
3			77868.42	062.66	121.42	315.65	615.07	809.32	376.84	571.36
4			624.36	77872.94		149.88	419.07	667.97	235.33	
5			332.88*	632.48			186.59*		070.81	364.03*
6			76996.70	339.49			71921.60		67887.55	
7									679.26*	

*Blended line.

TABLE I (Continued)
Vacuum wave numbers of the lines of the Lyman bands of H₂

8-11 band $\nu_0 = 66903.0_6 \text{ cm}^{-1}$		8-12 band $\nu_0 = 65488.0_8 \text{ cm}^{-1}$		8-13 band $\nu_0 = 64438.9_0 \text{ cm}^{-1}$		8-14 band $\nu_0 = 63816.9_4 \text{ cm}^{-1}$		9-6 band $\nu_0 = 79337.0_2 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		66930.81		65515.74			63844.69		79363.91	
1	66852.12	936.15	65446.04	530.06*			63799.52	883.41	79252.98	333.53
2	779.76	919.32	391.95*	530.06*			794.82	934.29	112.64	
3	687.34	881.62		520.85		64536.04	806.06	64000.36	78917.53*	104.37
4	577.25			501.87	64313.92		839.36	087.52*	670.65	
5	452.56			480.42					374.26	
6	317.71									

9-7 band $\nu_0 = 76554.8_4 \text{ cm}^{-1}$		9-8 band $\nu_0 = 74011.7_0 \text{ cm}^{-1}$		9-9 band $\nu_0 = 71718.7_4 \text{ cm}^{-1}$		9-10 band $\nu_0 = 69692.4_8 \text{ cm}^{-1}$		9-11 band $\nu_0 = 67955.8_2 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		76581.90		74038.72		71745.41	69719.22		67982.52	
1	76476.56	557.06	73939.51	020.09	71653.13	733.79*	69633.51*	714.38	67905.02*	985.57
2	347.98	482.09	822.81*	73957.12	549.85	683.59	545.04	678.42	831.51	965.40
3	170.78	357.36	664.28	850.86*	410.50	596.98	426.20	613.04	736.80	923.43
4	75947.33		464.97	703.19	237.10				623.43	861.76
5	679.70		228.14	516.70	032.76				494.37	782.87
6	372.19				70800.69				353.46	
7	028.10				544.37					
8					264.72					

9-12 band $\nu_0 = 66540.8_4 \text{ cm}^{-1}$		9-13 band $\nu_0 = 65491.6_8 \text{ cm}^{-1}$		9-14 band $\nu_0 = 64869.7_0 \text{ cm}^{-1}$		10-6 band $\nu_0 = 80359.2_4 \text{ cm}^{-1}$		10-7 band $\nu_0 = 77577.0_6 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		66567.30		65518.26		64896.36	80385.82		77603.33	
1	66498.85	579.36*	65460.43	541.00	64852.32	932.68	80275.22	354.02	77498.79	577.62
2	443.33		426.57	560.61	846.53	980.40	134.26		369.68	500.98
3	375.85	562.01	391.95*	578.59	855.47	65042.00	79938.33		191.28	376.72
4	299.16		359.97	598.26	885.46	123.69			76966.02	195.79
5	216.68	505.08	334.93	623.48	956.20	244.87			699.02	76952.00
6	132.48								382.24	
7	051.88									

10-8 band $\nu_0 = 75033.9_2 \text{ cm}^{-1}$		10-9 band $\nu_0 = 72740.9_6 \text{ cm}^{-1}$		10-10 band $\nu_0 = 70714.7_0 \text{ cm}^{-1}$		10-11 band $\nu_0 = 68978.0_4 \text{ cm}^{-1}$		10-12 band $\nu_0 = 67563.0_6 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		75060.17		72767.18			69004.20		67589.05	
1	74961.66	040.63	72675.43	754.28			68927.19	006.09*	67521.09	599.97
2	845.08	74976.15	571.52	702.98			853.47	68984.48	465.02	596.38
3	684.95	870.30	431.02	615.85	70446.94		757.35	942.80	396.45	582.06
4		712.74	256.48				642.55		318.34	
5			052.22		131.85		513.22*		236.08	

*Blended line.

TABLE I (Concluded)
Vacuum wave numbers of the lines of the Lyman bands of H₂

10-13 band $\nu_0 = 66513.8_s \text{ cm}^{-1}$		10-14 band $\nu_0 = 65891.9_2 \text{ cm}^{-1}$		11-7 band $\nu_0 = 78569.1_3 \text{ cm}^{-1}$		11-8 band $\nu_0 = 76025.9_9 \text{ cm}^{-1}$		11-9 band $\nu_0 = 73733.0_s \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		66539.83		65918.00		78594.16				73758.00
1	66482.61	561.53	65874.59	953.59*	78491.62	566.00*	75954.00		73667.30	742.56
2	448.18	579.36*	868.14	999.76	360.59	485.51	835.72	75960.64	562.68*	687.26
3	412.50	597.96	875.93	66061.41	179.80	354.14			419.26	593.59
4	379.10		904.64*		77950.57	173.54			240.66	
5	354.29				676.42	77946.27*	224.84		029.47	302.18
6	328.29				359.99	672.42				
7	304.99				76992.10†					

11-10 band $\nu_0 = 71706.7_7 \text{ cm}^{-1}$		11-12 band $\nu_0 = 68555.1_3 \text{ cm}^{-1}$		11-13 band $\nu_0 = 67505.9_5 \text{ cm}^{-1}$		11-14 band $\nu_0 = 66883.9_9 \text{ cm}^{-1}$		12-7 band $\nu_0 = 79531.5_0 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		71731.95				67530.83		66908.84		79556.56
1	71648.41*	723.32	68513.11*	68588.38	67474.78	549.95	66866.56	941.79	79453.57	529.13
2	557.16	682.14		580.90*	439.16	563.71*	858.96	981.04	323.09	
3	435.36	609.64	384.92	559.27	400.98	575.25	864.34	67038.34	142.70	293.46
4	285.12		302.64		363.65		889.08	111.80		
5	109.15		213.35				925.38			

12-8 band $\nu_0 = 76988.3_6 \text{ cm}^{-1}$		12-9 band $\nu_0 = 74695.4_0 \text{ cm}^{-1}$		12-10 band $\nu_0 = 72669.1_4 \text{ cm}^{-1}$		12-11 band $\nu_0 = 70932.4_8 \text{ cm}^{-1}$		12-12 band $\nu_0 = 69517.5_0 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		77013.30		74720.15		72694.19		70957.44*		
1	76916.27	76992.10*	74629.90		72610.67	686.33	70881.73	957.44*	69476.13	69551.23
2	797.97	929.86	525.61	656.59	519.68	651.33	806.52			
3	636.11	786.93*	382.02	532.59	398.31	548.95	708.66	859.33		498.92
4	437.90		209.34		254.49					
5	164.10		73968.65		048.40		430.27			

12-13 band $\nu_0 = 68468.3_2 \text{ cm}^{-1}$		12-11 band $\nu_0 = 67846.3_5 \text{ cm}^{-1}$		13-7 band $\nu_0 = 80464.9_4 \text{ cm}^{-1}$		13-8 band $\nu_0 = 77921.8_0 \text{ cm}^{-1}$		13-9 band $\nu_0 = 75628.8_4 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0						77946.27*				
1	68437.24*	68513.22*		67905.02*		80457.32	77849.84	920.13	75563.23	
2	401.00*							847.21		75572.55
3	364.03*	514.16*	67827.33	977.97	80070.90		563.58*	726.27*	310.55	
4	332.87						354.95	563.58*		
5	270.01						104.78	351.62		
6							76817.47†			

13-10 band $\nu_0 = 73602.5_3 \text{ cm}^{-1}$		13-11 band $\nu_0 = 71865.9_2 \text{ cm}^{-1}$		13-12 band $\nu_0 = 70450.9_4 \text{ cm}^{-1}$		13-14 band $\nu_0 = 68779.8_0 \text{ cm}^{-1}$		
<i>J</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>	<i>P(J)</i>	<i>R(J)</i>
0		73625.73*		71889.53*			68803.24	
1	73544.04	614.32	71815.19	885.56		68762.59	832.96	
2	451.36	568.30	737.96*	855.33		753.33	870.71*	
3	326.35	489.62	636.78	800.10	70276.40	755.57	918.83	
4	171.22	379.65	513.43			775.45		
5	72989.19	242.86	371.19		093.39			

*Blended line.

†Doubtful.

TABLE II

Int.	ν_{vac} , cm ⁻¹	Int.	ν_{vac} , cm ⁻¹	Int.	ν_{vac} , cm ⁻¹	Int.	ν_{vac} , cm ⁻¹	Int.	ν_{vac} , cm ⁻¹	Int.	ν_{vac} , cm ⁻¹	Int.	ν_{vac} , cm ⁻¹
1	59360.51	3	61328.10	0	62448.66	0	63406.27	00	64009.76	00	65142.90	00	65983.88
00	411.47	2	351.48	0	507.51	00	430.79	00	026.90	00	143.83	00	989.26
00	684.89	3	357.80	0	806.28	00	441.62	3	048.45	00	217.72	0	60016.60
00	60069.64	00	475.46	00	820.44	00	456.72	2	110.59	0	264.57	00	022.22
1	339.09	1	498.99	0	875.51	2	585.03	2	121.16	00	280.93	00	028.08
1	518.41	2	505.17	00	892.58	00	603.35	0	139.07	00	331.57	00	105.98
00	552.95	00	509.29	00	897.65	00	624.11	1	269.07	0	333.17	00	124.56
0	615.83	00	518.91	1	902.93	1	645.25	00	279.33	00	442.62	00	159.71
2	837.85	2	528.00	00	905.14	00	698.84	0	287.39	00	512.72	00	281.95
00	856.58	0	643.69	1	952.75	1	702.74	2	293.32	2	545.47	00	202.84
2	866.25	0	724.42	1	980.40	0	708.72	0	312.53	0	583.03	00	309.75
0	887.21	0	758.65	1	63063.85	1	729.65	0	383.98	0	618.14	0	409.18
3	911.79	00	771.29	00	084.90	00	736.47	0	480.89	00	627.96	0	430.33
1	980.62	1	800.22	0	090.18	00	738.80	00	551.44	0	633.57	2	531.89
00	61026.24	00	869.39	0	140.24	2	749.02	00	587.91	0	671.64	00	690.81
2	062.58	00	939.23	00	182.10	1	789.47	0	648.84	0	692.84	00	827.68
2	102.67	0	62021.55	0	240.00	3	811.55	0	712.92	00	781.24	00	906.35
1	156.48	00	086.98	00	271.83	00	816.53	00	718.35	00	803.02	0	67048.43
1	200.12	00	124.53	00	308.00	00	857.94	1	805.79	0	881.53	00	083.27
1	205.33	0	154.59	00	312.97	00	873.82	00	893.67	0	114.74	0	708.87
2	229.74	00	215.43	0	329.07	00	878.13	00	65004.87	00	889.15	0	116.86
2	251.24	00	335.57	00	334.20	00	893.92	00	029.01	00	927.84	1	144.70
0	271.30	00	352.19	0	352.99	2	905.30	00	060.30	00	961.08	1	159.78
1	278.03	00	368.92	00	380.03	00	984.44	1	062.41	00	964.48	0	166.91
1	307.27	0	371.11	00	385.27	00	992.61	00	134.31	00	972.60	00	225.82

Most of our assignments were checked by combination differences. The agreement of combination differences formed from unblended lines is within $\pm 0.1 \text{ cm}^{-1}$. Table III illustrates this agreement for the level $v'' = 11$. Similar tables were set up for all other vibrational levels of the upper and lower state but are not reproduced here in order to save space.

TABLE III
 $\Delta_2 F''(J)$ values derived from bands with $v'' = 11$

Band \ J	1	2	3	4	5	6
2-11	151.03	248.69	341.87	429.11		
3-11	(151.48*)	248.68*	341.95*	429.02	508.42	578.50
4-11	151.01	248.75*	341.93	429.12	508.52	(578.64)
5-11	151.00*	248.68	(342.18)	429.07	508.21	(578.35)
6-11	151.01	248.68	342.01	429.06	508.42	578.49
7-11	150.72*	248.69				
8-11	151.05	248.81	342.07	429.06*	508.30	
9-11	151.01	248.77	341.97	429.06		
10-11	150.73	248.74*	341.93	(429.58)		
12-11	150.92*	248.78*		429.06*		
13-11		248.78	341.90	428.91		
Mean	150.94	248.73	341.95	429.05	508.37	578.50

*From blended lines.

() Not included in the mean.

(a) *The Lower State, $X^1\Sigma_g^+$*

In Fig. 2 the values of $\{\Delta_2 F''(J)\} / \{4(J + \frac{1}{2})\}$ are plotted against $(J + \frac{1}{2})^2$ for all vibrational levels of the ground electronic state. As is well known, according to the relation

$$(1) \quad \frac{\Delta_2 F_v(J)}{4(J + \frac{1}{2})} = \left(B_v - \frac{3}{2} D_v + \frac{27}{16} H_v \right) - \left(2D_v - \frac{17}{2} H_v \right) (J + \frac{1}{2})^2 + 3H_v (J + \frac{1}{2})^4 \dots$$

the intercepts of the lines in Fig. 2 give the B_v values, the slopes the D_v values, and the curvatures the H_v values. For the actual determination of these quantities the plots were made for each v'' separately after addition of

$$a(J + \frac{1}{2})^2 - b(J + \frac{1}{2})^4$$

with suitable values of a and b to make the resulting line straight and of sufficiently small slope that both the remaining slope and the intercept could be determined accurately. The value of b is then $3H_v$ while the slope of the straight line added to a is $\{2D_v - (17/2)H_v\}$ and the intercept is $\{B_v - (3/2)D_v + (27/16)H_v\}$.

A check on the B_v and D_v values is obtained in the process of determining the vibrational differences $\Delta G''(v + \frac{1}{2})$. This was done by forming the differences of corresponding lines of bands with the same upper state and differing by $\Delta v'' = 1$ in the lower state according to the formula

$$(2) \quad \begin{aligned} R_{v'',}(J) - R_{v'',+1}(J) &= P_{v'',}(J) - P_{v'',+1}(J) = \Delta G''(v'' + \frac{1}{2}) \\ &\quad - (B_{v'',} - B_{v'',+1})J(J+1) + (D_{v'',} - D_{v'',+1})J^2(J+1)^2 \\ &\quad - (H_{v'',} - H_{v'',+1})J^3(J+1)^3 + \dots \end{aligned}$$

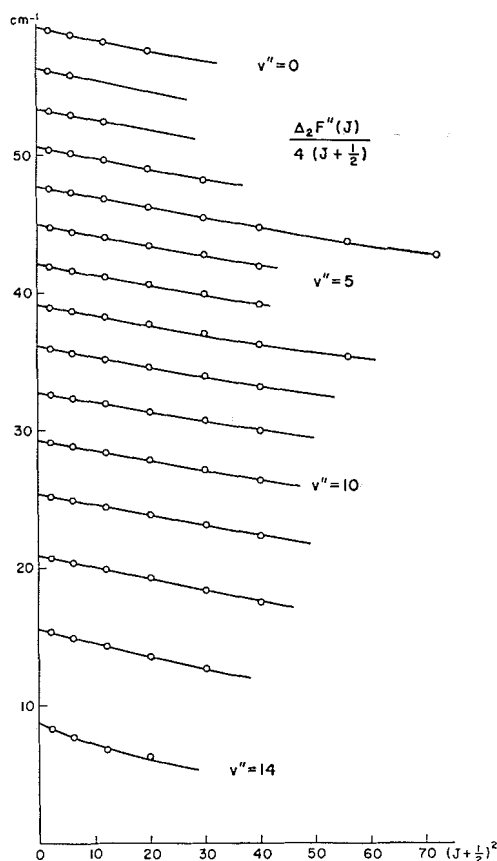


FIG. 2. Values of $\Delta^2 F''(J)/4(J+\frac{1}{2})$ for all observed vibrational levels of the ground state of H_2 .

If one adds to the numerical value of the left-hand side of this equation the numerical value of the quantity

$$(B_{v''} - B_{v''+1})J(J+1) - (D_{v''} - D_{v''+1})J^2(J+1)^2 + (H_{v''} - H_{v''+1})J^3(J+1)^3$$

with the previously determined B_v , D_v , and H_v values and plots against $J(J+1)$ one should obtain a horizontal line if the B_v , D_v , and H_v values were correct. The slope and curvature of the line actually obtained gives small corrections to these constants. Actually for hydrogen the terms in D_v and H_v are relatively much larger than for most other molecules and it is difficult to obtain them with sufficient accuracy from the plots of equation (2). In most cases, therefore, the D_v and H_v were taken from the $\Delta_2 F''$ values and only the B_v values were slightly adjusted. However, for $v'' = 13$ and 14 the number of $\Delta F''$ values available is insufficient and the constants were determined entirely from the ΔG plots. The curvature of these last ΔG plots is so large that it cannot be satisfactorily represented by (2). Therefore the B_{13} , D_{13} , B_{14} , D_{14} have much reduced accuracy.

In Table IV all the B_v'' , D_v'' , and H_v'' values thus obtained are collected. The constants for $v = 0, 1, 2$ are those of Stoicheff (1957), except that the values for $v = 1$ have been recalculated from his data using an approximate H_1 value from Fig. 6 below. The level $v = 3$ is the only one for which our measurements can be compared with Stoicheff's values. Stoicheff found $B_3 = 50.628_2$ and $D_3 = 0.0412_5 \text{ cm}^{-1}$ while we find $B_3 = 50.623_2$ and $D_3 = 0.0406_9$. The agreement is quite satisfactory. In Table IV the average of these two sets is listed for $v = 3$. For comparison with the new values the earlier B_v data of Jeppesen are listed in column 3. Experimental D_v values had previously been determined only for $v = 0, 2, 3$ while H_v had only been determined for $v = 0$ (Stoicheff 1957). In spite of the lower wave number accuracy the present data yield D_v and H_v of an accuracy comparable to the Raman work since they extend to higher J values.

TABLE IV
Rotational constants of the ground state of H_2

v''	B_v'', cm^{-1}		D_v'', cm^{-1}	$H_v'', \times 10^{-5} \text{cm}^{-1}$
	This paper	Jeppesen		
0	59.3392 ^a	59.373	0.0459 ₉ ^a	5.2 ^a
1	56.3685 ^b	56.391	0.0437 ₀ ^b	
2	53.4754 ^a	53.508	0.0427 ₇ ^a	
3	50.6257 ^d	50.676	0.0409 ₇ ^d	3.75
4	47.801 ₃	47.850	0.0397 ₁	3.50
5	44.958 ₁	44.954	0.0385 ₁	3.4 ₂
6	42.096 ₁	42.075	0.0377 ₉	3.3 ₃
7	39.145 ₃	39.116	0.0367 ₇	3.2 ₅
8	36.088 ₂	36.068	0.0365 ₃	3.2 ₀
9	32.833 ₀	32.694	0.0363 ₈	3.0 ₉
10	29.311 ₂		0.0366 ₀	2.5 ₀
11	25.399 ₁		0.0378 ₁	1.8 ₃
12	20.926 ₅		0.0413 ₀	(1.8 ₃) ^c
13	15.582 ₈ ^c		0.0492 ₇	—
14	8.770 ₁ ^c		0.0846 ₇	—

^aStoicheff's values.

^bObtained from Stoicheff's data using an interpolated H'' .

^cObtained from $\Delta G''$ plots.

^dAverage of Stoicheff's and our values.

^eAssumed value.

The $\Delta G''$ values obtained from the intercepts of the plots of equation (2) are listed in Table V. The first three values are from Stoicheff. The values of Jeppesen are given in column 3. The fourth column gives $\Delta_2 G''(v+1)$ and the fifth column $G_0(v)$ obtained by simply adding up the values in the first column.

In Figs. 3a, 4, 5a, 6, 7a, 8 are plotted the values of B_v , ΔB_v , D_v , H_v , $\Delta G(v+\frac{1}{2})$, and $\Delta^2 G(v+1)$. The B_v and ΔG_v curves have very similar behavior showing an initial slight positive curvature and toward the end a rapid convergence to zero. The D_v curve is of a quite different type reaching very high values at the last v'' values. H_v seems to change sign near $v = 12$.

The B_v and $\Delta G(v+\frac{1}{2})$ values are much more accurate than can be represented by plots like Figs. 3a and 7a. Even after subtraction of linear terms

TABLE V
Vibrational differences and vibrational levels in the ground state of H₂

$v + \frac{1}{2}$	$\Delta G''(v + \frac{1}{2})$		$\Delta^2 G''$		$G_0(v)$	v
	This paper	Jeppesen	This paper	This paper		
				0		0
$\frac{1}{2}$	4161.13 ₇ *	4160.96				
$1\frac{1}{2}$	3925.97 ₅ *	3922.18	235.16 ₂	4161.14		1
$2\frac{1}{2}$	3695.24 ₀ *	3693.82	230.73 ₅	8087.11		2
$3\frac{1}{2}$	3468.01	3467.82	227.23 ₀	11782.35		3
$4\frac{1}{2}$	3241.56	3239.51	226.45	15250.36		4
$5\frac{1}{2}$	3013.73	3013.96	227.83	18491.92		5
$6\frac{1}{2}$	2782.18	2782.01	231.54 ₅	21505.65		6
$7\frac{1}{2}$	2543.14	2543.36	239.04 ₅	24287.83		7
$8\frac{1}{2}$	2292.96	2292.55	250.18	26830.97		8
$9\frac{1}{2}$	2026.26	2026.28	266.70	29123.93		9
$10\frac{1}{2}$	1736.66	1737.53	289.60	31150.19		10
$11\frac{1}{2}$	1414.98	1416.92	321.68	32886.85		11
$12\frac{1}{2}$	1049.18		365.80	34301.83		12
$13\frac{1}{2}$	621.96		427.22	35351.01		13
				35972.97		14

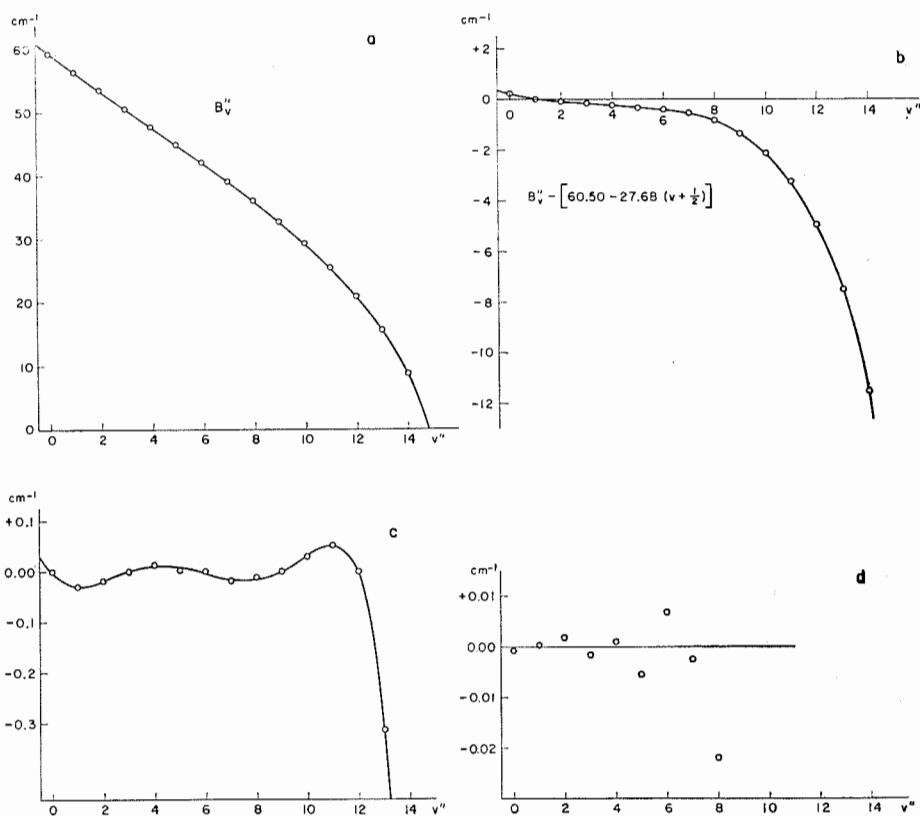
*Stoicheff's values.

as in Figs. 3b and 7b a very smooth curve remains which in each case shows a point of inflection at about $v'' = 3$. The rise of the first few points above the straight line representing the next few points is quite striking (see the initial drop of the ΔB and $\Delta^2 G$ curves) and apparently must be related to a peculiarity of the potential energy curve. A similar point of inflection has been observed for the ΔG values of the ground state of the O₂ molecule (Curry and Herzberg 1934). While H₂ and O₂ are the only cases of homonuclear molecules for which this anomaly has been observed in the ground state, it is quite common for polar molecules, for example HF, HCl, and others. It may be due to the contribution of ionic states to the ground state, since for ionic states a ΔG curve with positive curvature throughout is expected.

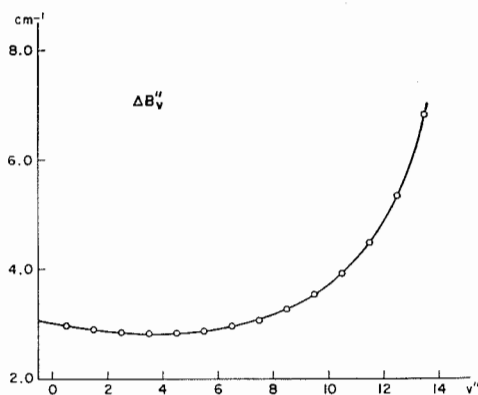
On account of the initial inflection it is difficult to represent the whole B_v'' and $\Delta G''$ curves by polynomial expressions except by using a very large number of terms. This is shown by Figs. 3c and 7c where the deviations from the five-term formulae

$$(3) \quad B_v'' = 60.818_6 - 2.9621_1(v + \frac{1}{2}) + 0.00458(v + \frac{1}{2})^2 \\ + 0.004548(v + \frac{1}{2})^3 - 0.0005105(v + \frac{1}{2})^4,$$

$$(4) \quad \Delta G''(v + \frac{1}{2}) = 4278.70_6 - 235.4967(v + \frac{1}{2}) + 0.5886_1(v + \frac{1}{2})^2 \\ + 0.2803_6(v + \frac{1}{2})^3 - 0.03790_3(v + \frac{1}{2})^4$$

FIG. 3. B_v curve for the $X\ ^1\Sigma_g^+$ ground state of H_2 .

- (a) Plotted directly.
 (b) Plotted after subtraction of a linear term.
 (c) Residuals of equation (3).
 (d) Residuals of equation (9).

FIG. 4. ΔB_v curve for the $X\ ^1\Sigma_g^+$ ground state of H_2 .

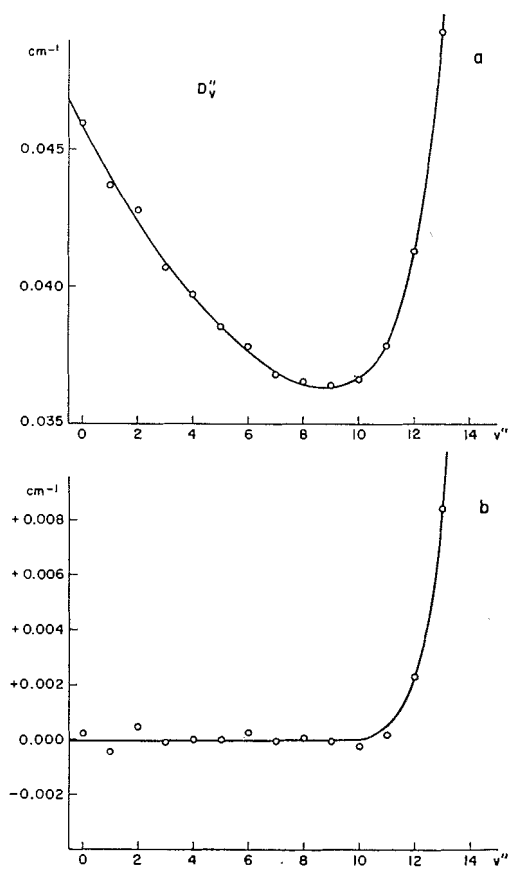


FIG. 5. D_v curve for the $X^1\Sigma_g^+$ ground state of H_2 .
 (a) Plotted directly.
 (b) Residuals of equation (15).

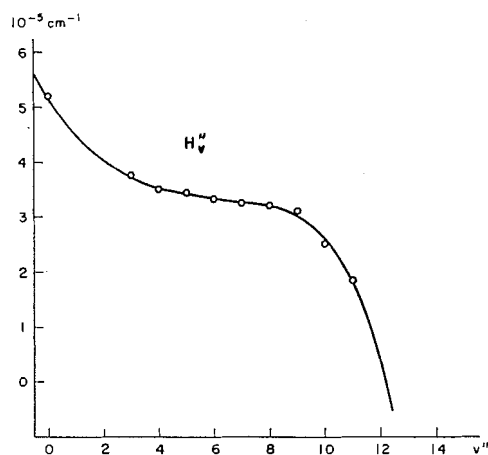


FIG. 6. H_v curve for the $X^1\Sigma_g^+$ ground state of H_2 .

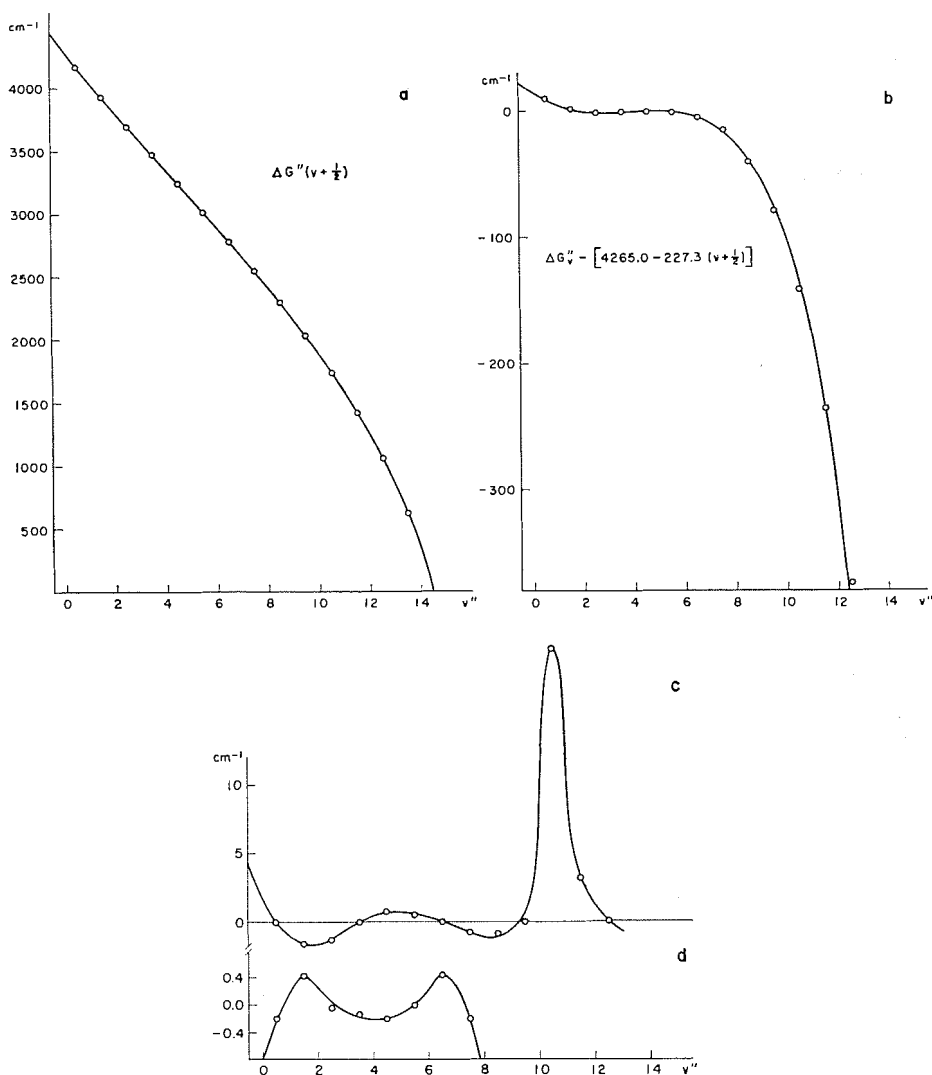


FIG. 7. $\Delta G(v + \frac{1}{2})$ curve for the $X^1\Sigma_g^+$ ground state of H.

- (a) Plotted directly.
- (b) Plotted after subtraction of a linear term.
- (c) Residuals of equation (4).
- (d) Residuals of equation (14).

are shown. These formulae were so chosen that the observed B_v and $\Delta G(v + \frac{1}{2})$ for $v = 0, 3, 6, 9, 12$ are exactly represented. It is seen that there are large systematic deviations which are much larger than the possible error of the individual B_v and $\Delta G(v + \frac{1}{2})$ values and which cannot be reduced by a different choice of coefficients in the equations. It is clear from Figs. 3c and 7c that the first term in (3) and (4) does not give the correct B_e and ω_e respectively. Rather,

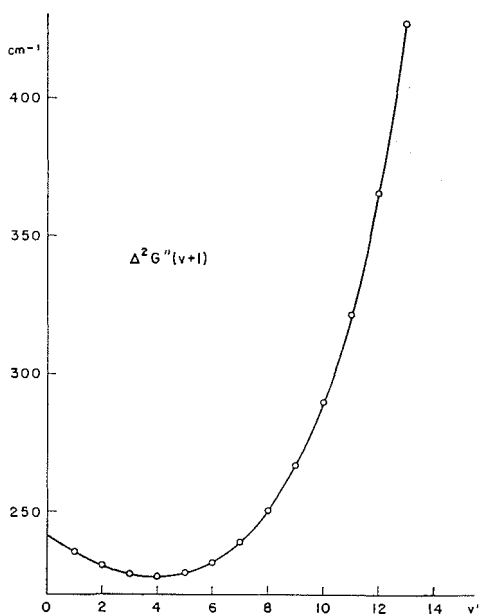


FIG. 8. $\Delta^2 G''(v+1)$ curve for the $X^1\Sigma_g^+$ ground state of H_2 .

in order to obtain these constants the curves in Figs. 3c and 7c have to be extrapolated to $-\frac{1}{2}$. Such an extrapolation is inherently beset with difficulties when the B_v and ΔG values do not follow a simple formula and, therefore, the B_v and ω_e values derived in the following discussion have appreciably larger limits of error than the individual B_v and ΔG values.

Stoicheff (1957) obtained from his B_0, B_1, B_2, B_3 values, by the method of least squares, the three-term formula

$$(5) \quad B_v'' = 60.840_7 - 3.0177_4(v + \frac{1}{2}) + 0.0285_5(v + \frac{1}{2})^2$$

which, when the slightly revised B_1 and B_3 values of Table IV are used, changes to

$$(6) \quad B_v'' = 60.842_1 - 3.0243_6(v + \frac{1}{2}) + 0.0302_5(v + \frac{1}{2})^2.$$

If the same B_v values are represented by a four-term formula one obtains

$$(7) \quad B_v'' = 60.864_3 - 3.0810_8(v + \frac{1}{2}) + 0.0644_5(v + \frac{1}{2})^2 - 0.0057_0(v + \frac{1}{2})^3,$$

while a five-term formula for the first five B_v values is

$$(8) \quad B_v'' = 60.868_7 - 3.0958_3(v + \frac{1}{2}) + 0.0788_7(v + \frac{1}{2})^2 - 0.0110_7(v + \frac{1}{2})^3 + 0.0006_7(v + \frac{1}{2})^4.$$

Finally a least squares four-term formula was developed from the first eight B_v values ($B_0 \dots B_7$):

$$(9) \quad B_v'' = 60.863_5 - 3.0763_8(v + \frac{1}{2}) + 0.0601_7(v + \frac{1}{2})^2 - 0.0048_1(v + \frac{1}{2})^3.$$

As shown in Fig. 3*d*, the residuals for the first five B_e values are less than 0.002 cm^{-1} . This together with the agreement with B_e from (7) and (8) suggests that a fairly satisfactory extrapolation to $-\frac{1}{2}$ has been obtained and that we can put $B_e = 60.864$, $\alpha_e = 3.076_4$, $\gamma_e = 0.0601_7$, $\delta_e = 0.0048_1 \text{ cm}^{-1}$. These constants are significantly different from Stoicheff's values which were based on a three-term formula. It is believed that the new B_e value represents an improvement.

With regard to the vibrational levels the situation is not as satisfactory. Using the first three, four, five, and six ΔG values one obtains the following three-, four-, five-, and six-term formulae (the first already given by Stoicheff)

$$(10) \quad \Delta G''(v+\frac{1}{2}) = 4280.299 - 239.457_0(v+\frac{1}{2}) + 2.1725_7(v+\frac{1}{2})^2,$$

$$(11) \quad \Delta G''(v+\frac{1}{2}) = 4280.666 - 240.4725_8(v+\frac{1}{2}) + 2.9050_0(v+\frac{1}{2})^2 \\ - 0.1536_7(v+\frac{1}{2})^3,$$

$$(12) \quad \Delta G''(v+\frac{1}{2}) = 4280.173 - 238.8198_3(v+\frac{1}{2}) + 1.2898_1(v+\frac{1}{2})^2 \\ + 0.4473_3(v+\frac{1}{2})^3 - 0.07512_5(v+\frac{1}{2})^4,$$

$$(13) \quad \Delta G''(v+\frac{1}{2}) = 4279.590 - 236.7367(v+\frac{1}{2}) - 1.0535_2(v+\frac{1}{2})^2 \\ + 1.5820_0(v+\frac{1}{2})^3 - 0.32179_2(v+\frac{1}{2})^4 + 0.019733(v+\frac{1}{2})^5.$$

There is no clear convergence. If one uses the first eight ΔG values for a four-term least square solution one finds

$$(14) \quad \Delta G''(v+\frac{1}{2}) = 4282.041_7 - 243.5764_0(v+\frac{1}{2}) + 4.5417_4(v+\frac{1}{2})^2 \\ - 0.3966_7(v+\frac{1}{2})^3.$$

The residuals from this formula are plotted in Fig. 7*d*. The deviation of $\Delta G(1\frac{1}{2})$ and $\Delta G(6\frac{1}{2})$ from the formula is far larger than any conceivable error of the observed values. One must assume that the true ΔG curve is strongly curved with respect to eq. (14) (see Fig. 7*c*) and therefore ω_e , $\omega_e x_e$, ... cannot be obtained from this formula. The ω_e values corresponding to the formulae (4), (10), (11), (12), (13), (14) are 4396.55, 4400.39, 4401.39, 4399.80, 4398.30, 4404.59 cm^{-1} respectively. Both the first and the last of these values are clearly not the true ω_e values since the residuals of (4) and (14) plotted in Figs. 7*c* and 7*d* indicate definitely a large deviation for $v = -\frac{1}{2}$. It is very difficult to determine which of the remaining four ω_e values (if any) is the correct one since there is no convergence. For lack of better information it seems best to retain Stoicheff's original value 4400.39 cm^{-1} based on the first three ΔG values since there is no good reason why the initial part of the curve should depend on the higher terms necessary to account for the higher ΔG values. One must, however, realize that while the ω_e value in itself is quite accurate, it depends strongly on the method of evaluation (i.e. the number of terms used). This large uncertainty of ω_e and the smaller one of B_e is unfortunate since it is the B_e , ω_e , ... values that can be most directly interpreted in terms of internuclear distances and vibrational frequencies and which are required if one wants to predict the constants of the isotopic molecules HD, D₂, HT, T₂, etc.

The D_v values can be represented by a polynomial more easily than the B_v and ΔG values. Figure 5b shows the residuals using the four-term expression

$$(15) \quad D_v'' = 0.04657_3 - 0.001616(v'' + \frac{1}{2}) - 1.43 \times 10^{-5}(v'' + \frac{1}{2})^2 + 7.6 \times 10^{-6}(v'' + \frac{1}{2})^3.$$

As shown by Figs. 3a and 7a, both the B_v and the ΔG curves are likely to go through zero between $v = 14$ and $v = 15$ in agreement with the fact that no bands with $v = 15$ were found in spite of considerable search. The breaking-off of the vibrational structure in the ground state of H_2 does not lead to a very accurate value for the dissociation energy D_0^0 since the last observed interval is still fairly large. However, it is consistent with the precise value $D_0^0 = 36116 \pm 6 \text{ cm}^{-1}$ obtained by Beutler (1935) from the onset of the far ultraviolet absorption continuum.

The energy of the $v'' = 14$ level above $v'' = 0$ (see Table V) is 35973.0 cm^{-1} . This is the $J = 0$ level. The $J = 1$ level lies $2B - 4D - 8H = 17.2 \text{ cm}^{-1}$ above this. Adding the observed $\Delta_2 F$ values successively to the energies of the $J = 0$ and $J = 1$ levels, one obtains the following values for the rotational levels of the last vibrational level

$J = 0$	35973.0,	$J = 1$	35990.2,	$J = 2$	36022.8,
$J = 3$	36067.6,	$J = 4$	36117.8,	$J = 5$	36153.4.

The last observed rotational level is definitely above the dissociation limit, while the second last is possibly above it. The existence of sharp rotational levels above the dissociation limit is easily understood since the effective potential energy curves for $J \neq 0$ have maxima which have to be passed before the molecule can dissociate (see e.g. Herzberg 1950b). Unfortunately it is not readily possible to predict the exact height of these maxima unless the potential curve for $J = 0$ were very precisely known for large r values. From the observation that lines with $J = 5$ are still quite sharp while lines with $J = 6$ are absent, presumably on account of extreme width, one must conclude that for barriers of more than 60 cm^{-1} the H atoms readily pass through. It will be interesting to compare the transmission of D atoms with those of H atoms through these barriers.

(b) *The Upper State, $B^1\Sigma_u^+$*

The molecular constants of the upper state were determined in substantially the same way as for the ground state. The combination differences $\Delta_2 F'$ as well as the $\Delta G'$ values agree very well with the values derived by Richardson and Davidson (1929) for the lower v values from an analysis of some visible bands of H_2 which have the $B^1\Sigma$ state as lower state. We have preferred not to mix our data with theirs but have evaluated B'_v , D'_v , $\Delta G'$ independently. On account of a number of perturbations the rotational constant H'_v could be determined only for a few levels. The average of these values $H'_v = 1.1 \times 10^{-5} \text{ cm}^{-1}$ was assumed to apply to all vibrational levels of the $B^1\Sigma$ state.

Our final values of B'_v and D'_v are given in Table VI and the values of $\Delta G'$ and $v_{00} + G'_0(v)$ in Table VII. The B_v values of Jeppesen (1933) which are

partially based on those of Richardson and Davidson (1929) are included in Table VI for comparison. Similarly the $\nu_{00}+G'(v)$ values given by Dieke (1958) and also based on Jeppesen's data are included in Table VII. The systematic almost constant deviations in the latter case (up to 9 cm^{-1}) are apparently due to lack of adequate standards in Jeppesen's work, which was

TABLE VI
Rotational constants of the upper state ($B\ ^1\Sigma_u^+$)

v	$B_v', \text{ cm}^{-1}$		$D_v', \text{ cm}^{-1}$
	This paper	Jeppesen	
0	19.453 ₇	19.449	0.0154 ₁
1	18.469 ₁	18.475	0.0133 ₁
2	17.653 ₈	17.646	0.0121 ₀
3	16.927 ₁	16.927	0.0111 ₁
4	16.265 ₃	16.260	0.0102 ₁
5	15.650 ₇	15.621	0.0094 ₅
6	15.066 ₆	15.000	0.0085 ₇
7	14.493 ₉	14.406	0.0076 ₃
8	14.027 ₂		0.0060 ₉
9	13.468 ₈		0.0058 ₇
10	13.165 ₂		0.0030 ₄
11	12.554 ₄		0.0042 ₁
12	12.610 ₃		—
13	11.797 ₂		0.0054 ₂

TABLE VII
Vibrational constants of the upper state ($B\ ^1\Sigma^+$)

$v+\frac{1}{2}$	$\Delta G' (v+\frac{1}{2}), \text{ cm}^{-1}$	$\nu_{00}+G'(v), \text{ cm}^{-1}$		v
		This paper	Dieke	
		90203.35	90195.46	0
$\frac{1}{2}$	1318.41			
		91521.76	91513.80	1
$1\frac{1}{2}$	1281.43			
		92803.19	92795.28	2
$2\frac{1}{2}$	1246.77			
		94049.96	94041.99	3
$3\frac{1}{2}$	1212.98			
		95262.94	95255.00	4
$4\frac{1}{2}$	1180.02			
		96442.96	96434.95	5
$5\frac{1}{2}$	1147.58			
		97590.54	97582.45	6
$6\frac{1}{2}$	1115.36			
		98705.90	98698.0	7
$7\frac{1}{2}$	1084.02			
		99789.92	99781.2	8
$8\frac{1}{2}$	1052.76			
		100842.68	100834.0	9
$9\frac{1}{2}$	1022.22			
		101864.90	101858.0	10
$10\frac{1}{2}$	992.07			
		102856.97	102852.3	11
$11\frac{1}{2}$	962.37			
		103819.34	103812.0	12
$12\frac{1}{2}$	933.44			
		104752.78	104751.3	13

particularly troublesome since he used a grazing incidence instrument. Jeppesen quotes an error of the relative λ values of $0.05 \text{ \AA}$ corresponding to 3 cm^{-1} in the region in which his work overlaps ours. Our wave numbers are believed to be free from such large systematic errors (see Section 2) and, therefore, the values of column 3 are considered to supersede those of Jeppesen.²

In Figs. 9a, 10, and 11a the B'_v , D'_v , and $\Delta G'$ values are plotted. It is seen that the shapes of the curves are very different from those of the ground state. There is a pronounced positive curvature for both B'_v and $\Delta G'$, much greater than the initial positive curvature in the ground state. The $B' \Sigma_u^+$ state is generally considered to be an ionic state and the behavior of the B'_v and $\Delta G'$ curves can be understood on this basis.

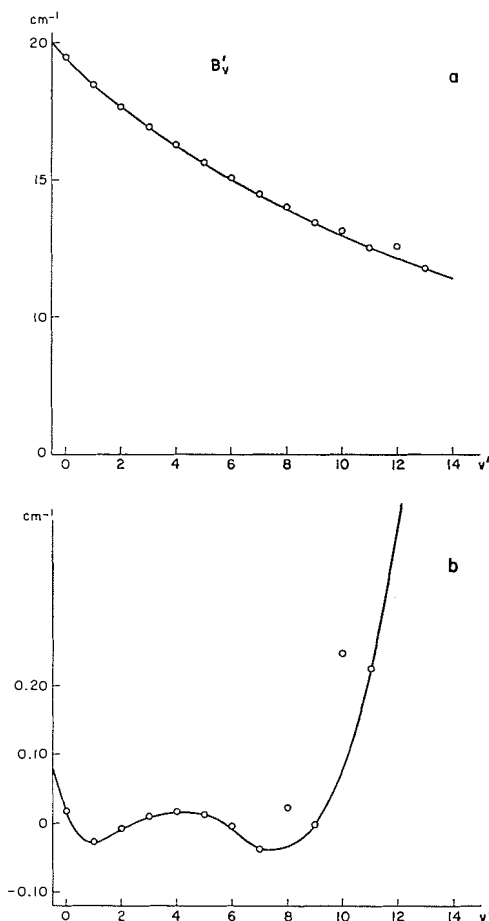
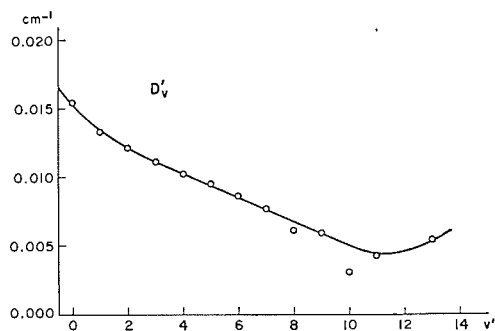
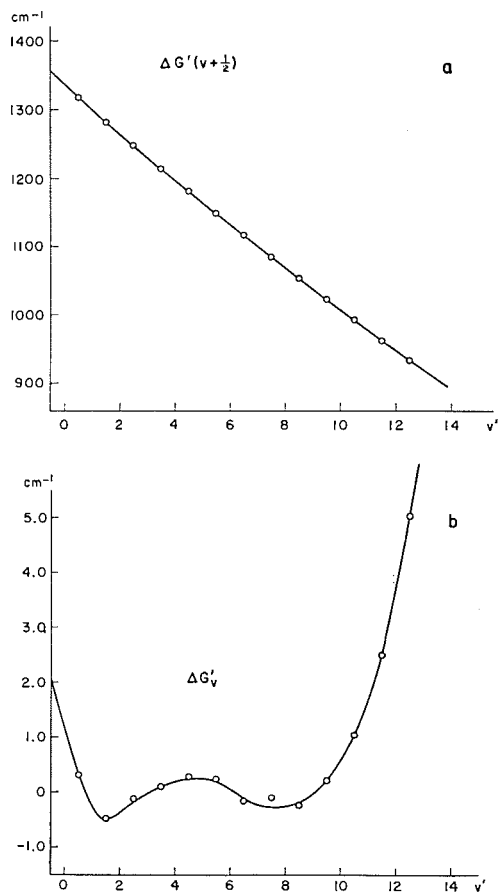


FIG. 9. B'_v curve for the $B' \Sigma_u^+$ upper state of the Lyman bands of H_2 .
(a) Plotted directly.
(b) Residuals of equation (18).

²It must be noted that Jeppesen's band origins ν_0 are larger by approximately $\frac{1}{4}(B'' - B')$ than Dieke's and ours since he uses the rotational term $F(J) = B(J + \frac{1}{2})^2$ instead of $BJ(J+1)$. His ν_0 unlike ours is not the wave number of the transition $J' = 0 \rightarrow J'' = 0$.

FIG. 10. D_v curve for the $B\ ^1\Sigma_u^+$ state of H_2 .FIG. 11. $\Delta G(v + \frac{1}{2})$ curve for the $B\ ^1\Sigma_u^+$ state of H_2 .
(a) Plotted directly.
(b) Residuals of equation (19).

As in the ground state, a large number of terms would be required to represent all the B' and $\Delta G'$ values within the accuracy of the measurements. The following two formulae represent the first four B'_v and $\Delta G'$ values exactly but the higher values increasingly poorly.

(16) $B'_v = 20.035 - 1.2312(v + \frac{1}{2}) + 0.14518(v + \frac{1}{2})^2 - 0.013450(v + \frac{1}{2})^3$

(17) $\Delta G'(v + \frac{1}{2}) = 1338.22 - 40.690(v + \frac{1}{2}) + 2.2475(v + \frac{1}{2})^2 - 0.24167(v + \frac{1}{2})^3$

while a least squares representation of the first 10 values yields

(18) $B'_v = 19.952 - 1.0619(v + \frac{1}{2}) + 0.06494(v + \frac{1}{2})^2 - 0.002628(v + \frac{1}{2})^3,$

(19) $\Delta G'(v + \frac{1}{2}) = 1336.68 - 37.503(v + \frac{1}{2}) + 0.6951_6(v + \frac{1}{2})^2 - 0.02464_0(v + \frac{1}{2})^3.$

There are considerable differences between the coefficients in these two sets of formulae. It would appear that the first set yields more significant values for the equilibrium constants $B_e, \alpha_e, \omega_e, \omega_e x_e \dots$ and these are therefore chosen as the final values in Table VIII. On the other hand the formulae (18) and (19) give a much better representation for the higher v' values. As is shown in Figs. 9*b* and 11*b*. Nevertheless there are systematic deviations which can be removed only by introducing still higher terms. Such a calculation did not seem worth while.

TABLE VIII

Molecular constants of H_2 in its ground state ($X^1\Sigma_g^+$) and first-excited state ($B^1\Sigma_u^+$)

	$X^1\Sigma_g^+$	$B^1\Sigma_u^+$	
B_0	59.339 ₂	19.453 ₇ cm ⁻¹	
B_e	60.864	20.035	
α_e	3.0763 ₈	1.2312	} from equations (8) and (14)
γ_e	0.0601 ₇	0.1451 ₈	
r_e	0.74158	1.29253	
r_e^*	0.74116		
D_e	0.04657	0.01625	} from equations (12) and (18)
β_e	-0.00161 ₆	-0.00216 ₈	
ν_{00}	0	90203.3 ₈	
T_e	0	91699.1 ₀	
ω_e	4400.39	1358.9 ₄	} from equations (10) and (17)
$\omega_e x_e$	120.81 ₈	21.52 ₉	
$\omega_e y_e$	0.724 ₂	0.8700 ₀	
$\omega_e z_e$		-0.0604 ₂	

The D'_v values (Tables VI and Fig. 10) can be represented by

(20) $D'_v = 0.01625 - 0.00216_8(v + \frac{1}{2}) + 2.28_9 \times 10^{-4}(v + \frac{1}{2})^2 - 1.18_5 \times 10^{-5}(v + \frac{1}{2})^3.$

Figures 9*b* and 10 show that the B'_v and D'_v values for $v' = 8, 10,$ and 12 are strongly perturbed. The higher vibrational levels of the $B^1\Sigma_u^+$ state starting with $v' = 8$ are in the same energy range as the lower vibrational levels of $C^1\Pi_u$. Indeed, if one plots the (unperturbed) total term values of the $B^1\Sigma_g$ and $C^1\Pi_u$ against $J(J+1)$ one finds intersections between $B^1\Sigma_g$ ($v = 8$) and $C^1\Pi_u$ ($v = 0$) at $J \sim 7$, between $B^1\Sigma_g$ ($v = 10$) and $C^1\Pi_u$ ($v = 1$) at $J \sim 6$, and between $B^1\Sigma_g$ ($v = 12$) and $C^1\Pi_u$ ($v = 2$) at $J \sim 4$, and no intersections for $B^1\Sigma, v = 7, 9, 11, 13$. These intersections of the (unperturbed) term

curves are clearly the reason for the anomalously high B and low D values of the levels with $v = 8, 10, 12$. Unfortunately no observations are available for high J values beyond the perturbations, and therefore it is not possible to obtain the true B'_v values for these levels.

The B'_v and $\Delta G'$ curves (Figs. 9b and 11b) also seem to indicate perturbations for $v' = 0$ or 1, or both, but no perturbing levels are located in this energy region. One must therefore conclude that the apparent irregularities are similar in nature to those found for the ground state at the lowest v values.

6. CONCLUSION

In Table VIII are collected the molecular constants derived from the analysis of the Lyman bands for the equilibrium positions of the upper and lower states. While these constants suffer from the difficulties mentioned earlier, it is believed that they represent improvements over those at present in the literature. It must be emphasized again that these constants represent well only the lower vibrational and rotational levels. For the higher levels the observed B_v , ΔG , and D values should be used.

In Table VIII two internuclear distances are given, one (r_e) based directly on the observed B_e and the other (r_e^*) after correction for various minor effects described more fully by Stoicheff (1957). With the new r_e'' value of H_2 the agreement of the r_e values of H_2 , HD, D_2 is no longer quite as good as according to Stoicheff. Improvements similar to the present ones for H_2 in the constants of D_2 , T_2 , and other isotopes appear desirable in order to find the cause of the apparent small discrepancies of the r_e values. Such investigations are being planned in this laboratory.

It would be interesting if theoretical work could be carried out which would give a more detailed explanation of the behavior of the B_v and ΔG curves for low v values. Such work would probably also enable one to derive more reliable equilibrium values, B_e and ω_e , and might remove the discrepancies in the r_e values.

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